

Coding & Solution

Date	07 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

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Smart Lender AI is a Machine Learning-powered web application developed using Python and Flask to predict loan approval based on applicant information. The solution integrates data preprocessing, feature scaling, Random Forest prediction, EMI calculation, applicant dashboard, PDF report generation, and an interactive user interface. The project follows modular programming practices with organized source code, reusable components, and cloud deployment through Render.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/kalyan-193/smart-lender
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, Bootstrap
Key Features Implemented	AI-based Loan Approval Prediction, Random Forest Model, Feature Scaling, Applicant Dashboard, Approval Probability, EMI Calculator, PDF Report Generation, Responsive UI, Dark/Light Mode, GitHub Integration, Render Deployment
Pending / Incomplete Features	Email notifications, Multi-user authentication, Database integration, Real-time banking API integration (Future Scope)
Setup / Run Instructions	Clone the repository → Create a virtual environment → Install dependencies using <code>pip install -r requirements.txt</code> → Run the application using <code>python app.py</code> → Open <code>http://127.0.0.1:5000</code> in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The Smart Lender AI application is designed using a modular architecture, separating the frontend, backend, Machine Learning model, and static resources into independent components for improved maintainability and scalability. The Random Forest model was selected after evaluating multiple Machine Learning algorithms based on prediction accuracy and cross-validation performance. The application provides an intuitive user experience through a responsive interface, interactive dashboard, EMI calculator, and downloadable PDF reports. The complete source code is maintained using Git and GitHub and deployed on Render for cloud-based access.